MAM3000W 3AL Solutions

University of Cape Town

Final Examination June 2022

Department of Mathematics and Applied Mathematics

Time: 120 minutes **Available marks:** 95 **Full marks:** 90

Instructions

1. Answer all questions in the lined answer booklets. **Start each question on a new page. Number each page and put your student number or EMPL ID on the top of each page.**
2. Show all work and reasoning unless otherwise instructed.
3. This exam consists of 5 questions. Make sure your question paper is complete.
4. If you've read the instructions write the name of your favourite mathematician on the inside cover of your booklet. You will receive one extra mark for doing so.

Good luck! ✧

1. (a) State the First Isomorphism Theorem.

   (b) Let $G = \mathrm{GL}_n(\mathbb{R})$ be the group of invertible $n \times n$ matrices over $\mathbb{R}$. We define a subgroup of G as follows:

$$S = \{A \in \mathrm{GL}_n(\mathbb{R}) : \det(A) = \pm 1\}.$$

   Prove that $G/S \cong \mathbb{R}^{>0}$, where $\mathbb{R}^{>0}$ is the multiplicative group of positive real numbers.

(a) **First Isomorphism Theorem.** *Let* $\varphi : G \to G'$ *be a group homomorphism. Then* $\ker \varphi \lhd G$ *and*

$$G/\ker \varphi \cong \operatorname{im} \varphi.$$

(b) Consider the map $\varphi : \mathrm{GL}_n(\mathbb{R}) \to \mathbb{R}^{>0}$ defined by $\varphi(A) = |\det(A)|$ for each $A \in \mathrm{GL}_n(R)$. For $A, B \in \mathrm{GL}_n(\mathbb{R})$, we have

$$\varphi(AB) = |\det(AB)| = |\det(A)\det(B)| = |\det(A)||\det(B)| = \varphi(A)\varphi(B).$$

Thus φ is a homomorphism. For each $x \in \mathbb{R}^{>0}$, we have $xI \in \mathrm{GL}_n(\mathbb{R})$. Moreover, $\varphi(xI) = x$. Thus φ is an epimorphism. Now, $A \in \ker \varphi$ if and only if $\varphi(A) = |\det(A)| = 1$. That is, $\det(A) = \pm 1$, or equivalently, $A \in S$. By the First Isomorphism Theorem, $\mathrm{GL}_n(\mathbb{R})/S \cong \mathbb{R}^{>0}$.

[5 + 10 = 15 marks]

2. (a) State the Third Isomorphism Theorem.

(b) Prove the Third Isomorphism Theorem.

(a) **Third Isomorphism Theorem.** *Let H and K be normal subgroups of G with $H \subseteq K$. Then $K/H \lhd G/H$ and*

$$(G/H)/(K/H) \cong G/K.$$

(b) Let us construct an epimorphism $\varphi : G/H \to G/K$ with $\ker\varphi = K/H$. We define $\varphi(gH) = gK$. Note that φ is well-defined since if $gH = g'H$ then $g^{-1}g' \in H \subseteq K$ and so $\varphi(gH) = gK = g'K = \varphi(g'H)$. Taking $gH, g'H \in G/H$, we have

$$\varphi((gH)(g'H)) = \varphi(gg'H) = gg'K = (gK)(g'K) = \varphi(gH)\varphi(g'H).$$

Moreover, for all $gK \in G/K$, we have $\varphi(gH) = gK$. Remember that the identity of a factor group G/K is K itself. Thus, $gH \in \ker\varphi$ if and only if $\varphi(gH) = gK = K$. That is, $g \in K$. If $g \in K$ then gH belongs to the factor group K/H. Thus, $\ker\varphi = K/H$. Finally, applying the First Isomorphism Theorem, we obtain $(G/H)/(K/H) \cong G/K$ as desired.

[5 + 15 = 20 marks]

3. Let G be a group. Recall that $\mathrm{Aut}(G)$ is the group of automorphisms of G (i.e. the set of isomorphisms $\varphi : G \to G$ equipped with the operation of composition). Consider the group action $* : \mathrm{Aut}(G) \times G \to G$ defined by $\varphi * x = \varphi(x)$.

   (a) Prove that $*$ is a group action.

   (b) Let $x \in G$. Prove that every element in the orbit of x has the same order as x.

(a) Identity: $\mathrm{id} * x = \mathrm{id}(x) = x$.

Compatibility: $\varphi * (\psi * x) = \varphi(\psi(x)) = (\varphi \circ \psi) * x$

(b) Suppose $y \in \mathrm{Aut}(G) * x$. Then there is some $\varphi \in \mathrm{Aut}(G)$ such that $\varphi * x = y$. That is, $\varphi(x) = y$. Since φ is an isomorphism, we then have $|x| = |y|$.

[5 + 10 = 15 marks]

4. Let G a group of order $p^n m$ where $n \geq 1$ and p is a prime such that $p \nmid m$. Suppose that H is a Sylow p-subgroup of G.

(a) Explain why H is a Sylow p-subgroup of $N_G(H)$ and why it is the *only* Sylow p-subgroup of $N_G(H)$.

(b) Let $a \in G$ such that $|a| = p^m$ where $1 \leq m \leq n$. Prove that if $aHa^{-1} = H$, then $a \in H$.

HINT: Every p-subgroup is contained in a Sylow p-subgroup.

(a) We have $|H| = p^n$. We know that H is a subgroup of $N_G(H)$. Thus $p^n \mid |N_G(H)|$. We know that $|N_G(H)| \mid p^n m$. Thus $|N_G(H)| = p^n k$ for some k where p does not divide k. Thus H is a Sylow p-subgroup of $N_G(H)$. Moreover, since H is normal in $N_G(H)$, it is the unique Sylow p-subgroup of $N_G(H)$.

(b) If $aHa^{-1} = H$, then $a \in N_G(H)$. Then $\langle a \rangle$ is a p-subgroup of $N_G(H)$ and so it is contained in a Sylow p-subgroup of $N_G(H)$. However, H is the only Sylow p-subgroup and a Sylow p-subgroup is a maximal p-subgroup. Therefore, $\langle a \rangle$ must be contained in H and so $a \in H$.

[10 + 10 = 20 marks]

5. (a) State the Third Sylow Theorem.

(b) State the definition of a simple group. Give an example of a group that is simple and an example of a group that is not simple.

(c) Show that any group of order $30 = 2 \times 3 \times 5$ is not simple.

(a) **Third Sylow Theorem.** *Let G be a group of order $p^n m$ where p is a prime that does not divide m. Let n_p denote the number of Sylow p-subgroups of G. Then $n_p \equiv 1 \bmod p$ and n_p divides m.*

(b) A group G is called **simple** if its only normal subgroups are G and $\{1\}$.

(c) By the Third Sylow Theorem, $n_5 \equiv 1 \bmod 5$ and $n_5 \mid 6$. Given the divisors of 6, we see that the only options are $n_5 = 1$ or 6. If $n_5 = 1$, there is a unique Sylow 5-subgroup and it is normal and so G is not simple and we are done. So, assume that $n_5 = 6$. Similarly, $n_3 \equiv 1 \bmod 3$ and $n_3 \mid 10$. So either $n_3 = 1$ or 10. Again, if $n_3 = 1$, we are done. So, assume $n_3 = 10$.

We are assuming $n_5 = 6$ and $n_3 = 10$. Our aim is to reach a contradiction, which will mean that either $n_5 = 1$ or $n_3 = 1$, and so G fails to be simple.

Let $P_1, \ldots, P_6$ be the Sylow 5-subgroups of G. By Lagrange's Theorem, the possible orders for the intersection of distinct Sylow 5-subgroups are 1 and 5. However, it cannot be 5 since the Sylow 5-subgroups are assumed to be distinct. Thus $P_i \cap P_j = \{1\}$ for $i \neq j$. Each Sylow 5-subgroup consists of 4 unique elements of order 5, not appearing in any other Sylow 5-subgroup. Thus there are $4 \times 6 = 24$ elements of G with order 5.

Through similar reasoning, there are $2 \times 10 = 20$ elements of G of order 3. This means there are at least $24 + 20 = 44$ elements in G. However, $|G| = 30 < 44$. So, it is impossible for both $n_5 \neq 1$ and $n_3 \neq 1$ and we are done.

[5 + 5 + 15 = 25 marks]